TYLER B. RANDALL

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EDUCATION

Bachelor of Science: Electrical Engineering (Expected 2028)
Brigham Young University, Provo, UT – Dean's List (Winter 2026 & 2025)
• **Overall GPA:** 3.94 / 4.00
• **Major GPA:** 3.97/4.00
• **Relevant Coursework:** Computer Systems, Circuits, Computer Science, Product Design & Development, Data Structures & Algorithms, Linear Algebra, Statistics for Engineers, Ordinary Differential Equations, Calculus I/II/III
• **Corner Canyon High School —** Valedictorian, 4.0 GPA

TECHNICAL SKILLS

• **Programming:** Python, SageMath, MATLAB, C/C++, Bash, LaTeX

• **Engineering & Electronics:** Arduino, PCB/SMD Assembly, Soldering, Breadboarding, Circuit Prototyping, Schematic Design

• **Software & Tools:** Git/GitHub, NGSpice , Autodesk Fusion 360, Eagle PCB, Blender

• **Computing:** Linux, High-Performance Computing (HPC)

• **Research:** Post-Quantum Cryptography, Technical Paper Writing, Technical Presentation

• **Language:** Fluent in Spanish

RESEARCH/TECHNICAL PROJECT EXPERIENCE

Research Assistant — BYU Dr. Harrison Lab (Post-Quantum Cryptography) | 2024–2026

• Implemented a Dual Concatenated McEliece-based cryptosystem in SageMath, including system instantiation and encryption of binary messages.

• Parallelized cryptanalytic attack functions for execution on BYU's supercomputer, dramatically reducing attack simulation runtime and enabling large-scale simulations.

• Co-authored two conference papers, earning 2nd place in both the graduate and undergraduate paper competitions at the International Telemetry Conference.

• Co-authored an additional manuscript intended for publication in the IEEE Transactions on Information Forensics and Security journal and wrote much of its simulation section.

• Investigated Quantum Error Correction and possible approaches using LDPC codes.

BYU Spacecraft Club | 2025–26

• **FEMTOSAT** — Helped create an Eagle schematic for a small PCB containing a microprocessor, radio, accelerometer, and barometer. Assisted with component placement and reflow soldering using a small oven. Launched in small rockets while recording pressure and temperature data; team won first place.

• **PocketQube** — Prototyped and tested battery-monitoring circuitry on breadboards using an Arduino before transitioning the design to an SMD PCB. Assisted with SMD component placement and field launch operations, including weather-balloon deployment.

BYU IMMERSE Program | 2025–26

• Continued research projects in cryptography, quantum error correction, and ethics in engineering.

BYU Rocketry Club | 2025

• Used OpenRocket simulations to support rocket design and assisted with physical assembly, including fins and body construction.

Singapore Product Design Study Abroad | 2026

• Led a five-person, multicultural team through user research and prototype development to address shoe storage and drying challenges in Singapore's rainy climate.

• Developed a working prototype capable of drying a pair of shoes using repurposed materials from the NUS campus, including cardboard and PVC pipe.

Research Assistant — BYU Dr. Chahal (Number Theory & Geometry) | 2025

• Investigating the history and mathematical proofs related to constructing regular polygons inscribed in circles.

OTHER WORK EXPERIENCE

Linear Algebra Grader | 2026

• Graded written math problems for Elementary Linear Algebra and offered feedback to students.

Grade Potential Tutor — Mathematics | 2025

• Tutored early college-level math students, strengthening conceptual understanding.

BYU Store Warehouse Prep Team | 2024

• Assisted with warehouse organization and preparing items for sale.

LEADERSHIP & HONORS

Masonic Foundation of Utah Scholarship | 2026–27
Clifford Aggen Award, 2nd Place — ITC Conference Undergraduate Paper Competition ($750) | 2026
BYU Engineering Leadership Certificate Program | 2025–28
College of Engineering Dean's List | Winter 2026 & 2025
Clifford Aggen Award, 2nd Place — ITC Conference Graduate Paper Competition ($1,000) | 2025
Mission Leadership (Zone Leader, District Leader, Trainer) — Bolivia | 2022–2024

SERVICE

Student Mentor in Math (Spanish) — Centennial Middle School | 2025–2026
Provo Food Bank Volunteer | 2025–26
Full-Time Missionary — Church of Jesus Christ of Latter-day Saints of Latter-day Saints, Bolivia | 2022–2024

• Completed humanitarian projects: built bathrooms, dug wells, and cleared terrain.